

5. MANUAL CALCULATION OF GAUSS-SEIDEL LOAD FLOW

GIVEN DATA

Line Impedances

$$Z_{12} = 0.02 + j0.04 \text{ p.u.}$$

$$Z_{13} = 0.01 + j0.03 \text{ p.u.}$$

$$Z_{23} = 0.0125 + j0.025 \text{ p.u.}$$

Slack Bus

$$V_1 = 1.05 \angle 0^\circ \text{ p.u.}$$

Load Data

Bus 2 :

$$P_2 = -2.566 \text{ p.u.}$$

$$Q_2 = -1.102 \text{ p.u.}$$

Bus 3 :

$$P_3 = -1.386 \text{ p.u.}$$

$$Q_3 = -0.452 \text{ p.u.}$$

Initial Voltages:

$$V_2^{(0)} = 1 + j0$$

$$V_3^{(0)} = 1 + j0$$

$$\text{Tolerance} = 10^{-6}$$

$$\text{Maximum number of iterations} = 100$$

SOLUTION

STEP 1 : CALCULATION OF LINE ADMITTANCES

$$Y_{12} = 1/(0.02 + j0.04)$$

$$= (0.02 - j0.04)/(0.02^2 + 0.04^2)$$

$$= 10 - j20 \text{ p.u.}$$

$$Y_{13} = 1/(0.01 + j0.03)$$

$$= (0.01 - j0.03)/(0.01^2 + 0.03^2)$$

$$= 10 - j30 \text{ p.u.}$$

$$\begin{aligned}
 Y_{23} &= 1/(0.0125+j0.025) \\
 &= (0.0125-j0.025)/(0.0125^2+0.025^2) \\
 &= 16-j32 \text{ p.u.}
 \end{aligned}$$

STEP 2 : FORMATION OF Y-BUS MATRIX

$$\begin{aligned}
 Y_{11} &= Y_{12} + Y_{13} \\
 &= 20-j50 \\
 Y_{22} &= Y_{12} + Y_{23} \\
 &= 26-j52 \\
 Y_{33} &= Y_{13} + Y_{23} \\
 &= 26-j62 \\
 Y_{12} &= Y_{21} = -10+j20 \\
 Y_{13} &= Y_{31} = -10+j30 \\
 Y_{23} &= Y_{32} = -16+j32
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 Y_{BUS} &= \\
 &\begin{bmatrix} 20-j50 & -10+j20 & -10+j30 \\ -10+j20 & 26-j52 & -16+j32 \\ -10+j30 & -16+j32 & 26-j62 \end{bmatrix}
 \end{aligned}$$

STEP 3 : GAUSS-SEIDEL ITERATIONS

Using

$$V_i = (1/Y_{ii})[(P_i - jQ_i)/V_i^* - \sum Y_{ij}V_j]$$

The voltages are updated iteratively.

The solution converges after 12 iterations.

FINAL BUS VOLTAGES

Bus 1

$$|V_1| = 1.0500 \text{ p.u.}$$

$$\text{Angle} = 0.0000^\circ$$

Bus 2

$$|V_2| = 0.9818 \text{ p.u.}$$

$$\text{Angle} = -3.5035^\circ$$

Bus 3

$$|V_3| = 1.0012 \text{ p.u.}$$

$$\text{Angle} = -2.8624^\circ$$

VOLTAGES IN RECTANGULAR FORM

$$V_1 = 1.0500 + 0.0000i$$

$$V_2 = 0.9800 - 0.0600i$$

$$V_3 = 1.0000 - 0.0500i$$